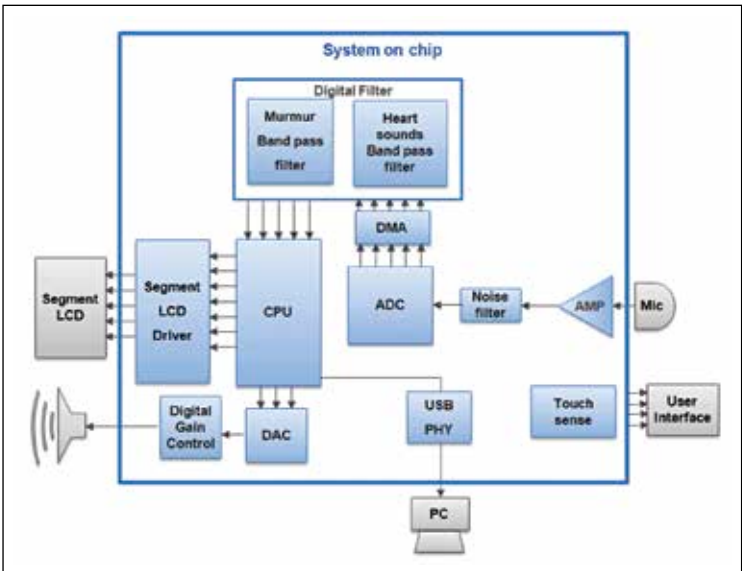


Architectural evolution

In the early 70s, high levels of integration were achieved. However, the main problem was the compactness and the cost of manufacturing such integrated circuits. The architecture was relatively simple back then with a basic CPU being attached to various peripherals on board. These components like Graphical Audio interfaces, modems, codes, DSPs etc. later evolved and themselves became processing units with their own on-board peripherals. During the 80s, the complexity and functionality of these components evolved and soon, constructors began devising ways to make them universally compatible to various microprocessors. Embedded systems or Application Specific Integrated Circuits (ASIC) became a reality in the late 80s.

There were rapid advances made subsequently in semiconductor processing technologies. Recent advents like the HDTV and smartphones show an increasingly evident need for incorporating the traditional microprocessor, peripherals and memory onto a single chip. With new and more complex devices coming in the near future, there is a need for adaptive SoCs.

There are certain factors that are to be taken into account when designing an efficient SoC -



The architecture of a phonocardiograph SoC